

Innlevering 2 (Kapittel 3 & 4)

Skrevet av: Justas Simutis, Dato: February 13, 2025

Oppgave 3.1

a) Creating a truth table of $(\neg\neg P \rightarrow P)$

P	$(\neg\neg P \rightarrow P)$
0	1
1	1

b) Creating a truth table of $\neg(P \rightarrow (Q \vee P))$

P	Q	$(Q \vee P)$	$\neg(P \rightarrow (Q \vee P))$
0	0	0	0
1	0	1	0
0	1	1	0
1	1	1	0

c) Creating a truth table of $(P \wedge (Q \vee R)) \rightarrow (P \vee (Q \wedge R))$

$F = (P \wedge (Q \vee R))$

$G = (P \vee (Q \wedge R))$

$H = (P \wedge (Q \vee R)) \rightarrow (P \vee (Q \wedge R))$

P	Q	R	$(Q \vee R)$	$(Q \wedge R)$	F	G	H
0	0	0	0	0	0	0	1
1	0	0	0	0	0	1	1
0	1	0	1	0	0	0	1
1	1	0	1	0	1	1	0
0	0	1	1	0	0	0	1
1	0	1	1	0	1	1	0
0	1	1	1	1	0	1	1
1	1	1	1	1	1	1	1

Oppgave 3.2

P.S. The hint is huge, couldn't have figured out that out of 6 there are 3 pairs. But for real though, I guess it doesn't hurt to include it.

Anyways, by looking at truth tables I found out the following relations:

1. $A \rightarrow B = \neg(\neg A \wedge B) = \neg A \vee B$
2. $\neg(A \wedge B) = \neg A \vee \neg B$

After that I tried to simplify everything into just OR and AND and see which ones are the same. Thus, I got:

- a) $\neg P \vee (Q \wedge R)$
- b) $\neg P \vee \neg Q \vee R$
- c) $P \vee \neg Q \vee R$
- d) $\neg P \vee (Q \wedge R)$
- e) $\neg P \vee \neg Q \vee R$
- f) $P \vee \neg Q \vee R$

Although after e) I didn't need to find f), I decided to verify the answer anyways. But due to an epic hint I got that was totally unnecessary since the last pair are the two not used ones for the last pair. Thus the result is:

1. a) = d) : $(P \rightarrow (Q \wedge R)) = \neg(P \wedge (\neg Q \vee \neg R))$
2. b) = e) : $((\neg P \vee \neg Q) \vee R) = ((P \wedge Q) \rightarrow R)$
3. c) = f) : $(P \vee (Q \rightarrow R)) = (\neg P \rightarrow (\neg Q \vee R))$

Oppgave 3.3

Honestly I can do something like this: $\neg P = \neg\neg\neg P$ and just repeat that, but where is the fun in that? So instead I will just apply some of the simple rules I did earlier. Rules such as $\neg(A \vee B) = \neg A \wedge \neg B$, $\neg(A \wedge B) = \neg A \vee \neg B$, and $A \rightarrow B = \neg A \vee B$. Thus $((\neg P \wedge \neg Q) \vee \neg R)$ becomes equivalent to:

1. $(\neg(P \vee Q) \vee \neg R)$
2. $\neg((P \vee Q) \wedge R)$
3. $(R \rightarrow (\neg P \wedge \neg Q))$
4. $(R \rightarrow \neg(P \vee Q))$ A bonus one :D

Oppgave 3.4

a) $(P \rightarrow Q) \vee R$. Explanation: IMPLIES is always True except for one, and OR is always True. So I just combined the two.

b) $\neg(\neg P \wedge Q \wedge R)$. Everything is True except for one thing, just like in an AND gate, we just switched False to True and where it starts. Simple enough.

Oppgave 4.1

a) $P \rightarrow Q$ only tells us what would happen if P were True. It itself does not assert P . However, from $Q \wedge P$ we assert both P and Q to be True. Thus they are not equivalent. For the same reason $Q \wedge P$ is the logical consequence of $P \rightarrow Q$.

b) This is a similar one to a). The logical consequence of $(P \rightarrow Q) \wedge (Q \rightarrow P)$ is obviously $P \wedge Q$ just as in a) because we can derive it from P and Q That we get from $P \wedge Q$. They are also not equivalent because $(P \rightarrow Q) \wedge (Q \rightarrow P)$ asserts both $P \rightarrow Q$ and $Q \rightarrow P$ - both of which do not assert P nor Q but the result of what would happen if one would be True.

Oppgave 4.2

- a) False, because it leaves the possibility of A being valid which may make A not falsifiable any longer.
- b) True. Tautological - Always True: Something that can not happen if A is falsifiable. Contradiction - Always False: Something that can not happen if A is satisfiable
- c) False. Example: $P \rightarrow Q$. We can not assert P .
- d) True. If a formula is always True no matter what premises are it doesn't change no matter what. This is a direct definition of validity.
- e) True. If a formula is always False we never need to check for the validity of whatever it is that follows.
- f) False. Due to contradictions tautologies can be derived from them.

Oppgave 4.3

- a) No. We can not deduce A . We can however deduce C and B .
- b) Yes. From $A \wedge C$ we know that both A and C must be True. We also know that if C then B , but also just B . Since we know that A and B we can safely deduce $A \wedge B$. I've had IN3070 before which is where I learned that myself and how to use LaTeX as well :D

Oppgave 4.4

- a) None. There doesn't exist a contradiction or a tautology. No $P \wedge \neg P$ or $P \vee \neg P$ when looking at it, also found an example to satisfy and falsify it.
- b) When using some of the rules I can easily see and identify the $\neg A \vee A$, same with B . So yeah... a tautology it is.
- c) Easily found an example of both True and False. Assume P and Q to be False. This makes it satisfiable. Now find example where either left or right is False, e.g. P is True and Q is False. So it is neither.
- d) We have had two that are neither, one that is valid, so this one is a contradiction. I am not joking, but the reason why is actually different. The thing is that left Must be True which is possible if and only if both A and B are True. However, the one on the left must also be True. That is not possible because if both A and B are True, the right part evaluates to False. Thus it can never be satisfied.